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A Appendix

A.1 Practical notes on evaluating Activation Oracles

While building AObench, we encountered several measurement issues that materially affect reported AO performance. We document them here because they apply to any AO evaluation and are easy to get wrong.

Use AUC, not accuracy, for binary classification tasks. Jakkli et al. [2026] reported near-chance AO performance on tasks such as sycophancy detection. We find that this is largely a calibration artifact: Qwen-based AOs frequently default to answering "No" regardless of the question, which makes fixed-threshold accuracy near-chance but leaves a strong signal in the difference between the "Yes" and "No" token logits. On a sycophancy-from-CoT task, the original AO scores 0.50 accuracy but 0.83 ROC AUC (Fig. 9). AUC is also markedly less sensitive to prompt phrasing than accuracy is; we recommend it as the default metric on yes/no AObench items.

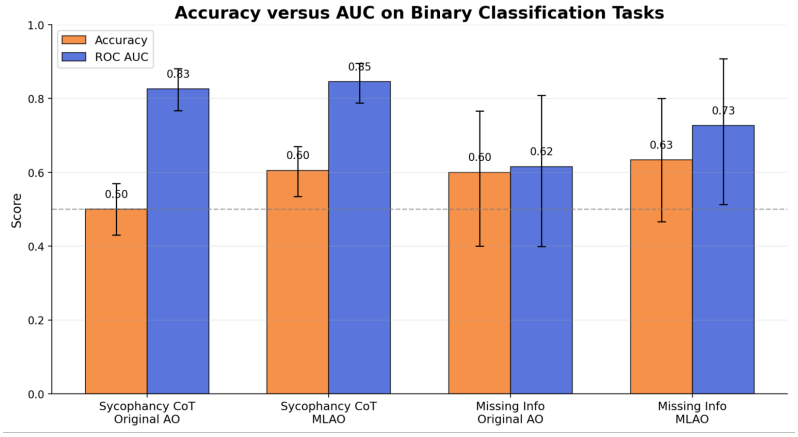


Figure 9: **Accuracy underestimates AO capability on Yes/No tasks.** Qwen-based AOs default to "No" on several binary-classification items, which suppresses accuracy without affecting the Yes/No logit margin. ROC AUC is much higher and more stable across phrasings.

Sweep the AO's context window when comparing to black-box baselines. Many open-ended AO tasks (e.g., "why is the model about to backtrack?") concern information spread across tens of tokens of internal computation. Restricting the AO to the final activation alone consequently produces misleadingly weak performance. On a Qwen3-8B backtracking task, the original AO scores $1.26/5$ mean correctness given only the final-token activation, but $2.10/5$ given the last 50 tokens, above the black-box baseline of asking Qwen3-8B the same question with full text context (Fig. 10). We therefore report AObench results at a context window of ≥ 20 tokens by default, and recommend the same when comparing AOs to text-only baselines.

Consensus sampling materially mitigates open-ended hallucination. For free-form AO answers, a simple inference-time strategy is to sample k completions at non-zero temperature and retain only those for which at least a fraction τ of samples agree. On the taboo secret-word task, unfiltered single-token accuracy is 46.6%; requiring consensus ≥ 0.8 over $k = 10$ samples retains 19.4% of items at 94.3% precision, with a smooth precision/recall trade-off (Fig. 11). This is a cheap, training-free mitigation we recommend deploying alongside any AO used to surface specific factual claims.

A.2 Our advice for training Activation Oracles

Our initial impression was that we could improve AOs by training on narrow tasks. Specifically, we singled out the tasks from "Test your best methods on our hard CoT interp tasks" (datasets can be found here). We found that we could quite consistently match the performance of linear probes when narrowly training, but never significantly exceeded it.

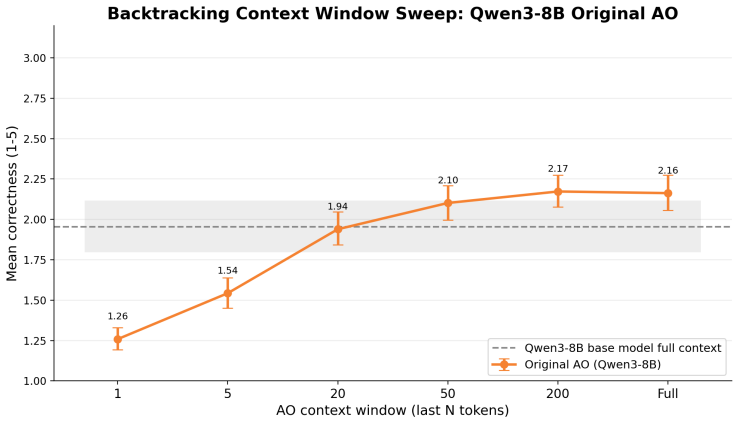


Figure 10: **Backtracking accuracy vs. AO context window.** Performance rises steeply with the number of activation positions supplied. The AO matches the black-box baseline at ∼20 tokens and exceeds it at 50.

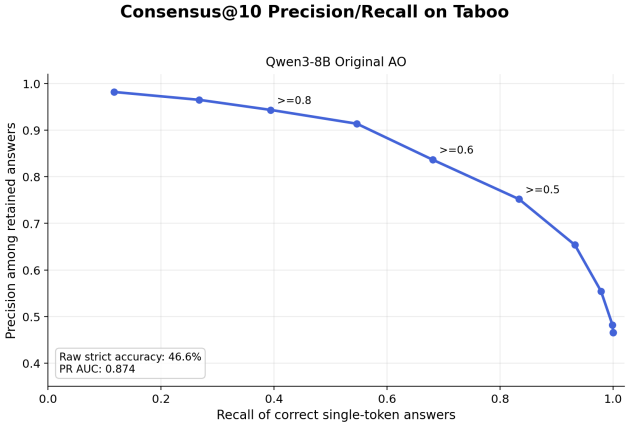


Figure 11: **Consensus@10 precision/recall on Taboo.** Requiring agreement among $k = 10$ samples cleanly trades coverage for precision, mitigating hallucination on the secret-word extraction task.

- Make a good eval, that you think a good Activation Oracle should be able to do (**solvability**) and is hard for a black box monitor (**text inversion**; you can explicitly check this)[5]. Then try to find training tasks that would make the oracle better at this.
- You should generally aim to match the performance of probes.[6]
- A good training task causes broad uplift, and is scalable.
- Loss graphs going down does not always translate to capability: in particular, future/pastlens demonstrates a very strong scaling law, but there is a risk of just fitting surface statistics that will not translate to any meaningful uplift in evals.
- We observed the majority of uplift on the evals after 10% of training (~200K tokens); training to convergence is generally not necessary to detect whether a task causes uplift.

[5]A thing we tried to do during our sprint, was training a model that does not get activations, but everything in the context window up until that point (the same data the activations get to attend to), trained on the same objective. The reason this might work is that, because AOs are the same model, this model would actually be able to internally access these same activations. We did run this baseline at one point early in our sprint, and performance was similar, but we are sufficiently uncertain that we do not feel comfortable sharing this result as strong evidence. If you were able to demonstrate this matches AO performance on all tasks, AOs would still be useful as an interpretability tool (because the statement "does this specific activation contain this information" is still interesting, or for auditing different models), but this would mean they are not more useful as a monitorability tool.

[6]It was a very consistent observation that we were able to match linear probe performance, but never significantly exceed it.

- Be careful when changing learning rate, LoRA rank or LoRA alpha, as they can destabilize training.
- We experimented with scheduling training tasks one by one (unshuffled) to locate uplift, but encountered catastrophic forgetting on tasks not included in the group. Therefore, we recommend you have at least 10% of data at every stage come from other tasks. An interesting way forward would be to have a broad "pre-training stage", say of verbatim and conversational data, and then a shorter "post-train" on specialized tasks.[7]
- Read your datasets, oracle outputs, and evaluation traces. Language models are not very good at generating or discriminating good AO questions/responses, so manual inspection helps verify the pipeline is behaving as intended.

Future directions we are excited about.

- Increasing corpus diversity on the unsupervised learning task.[8]
- Feeding even more, or all, layers and positions.
- Training directly on activations from finetuned models, to optimise for model-diffing tasks.

A.3 Feeding multiple positions

We observed that hallucinations are reduced by feeding activations from more positions to the AO. The link was approximately linear: the more activations are fed, the fewer hallucinations. This makes sense, since the relevant information may be spread out across many activations and need not be concentrated in any single one.

A.4 Experiments using post-training

We experimented with DPO after final training to improve on instruction-following, hallucination rate, and vagueness. Results were hard to stabilize, and we frequently ran into mode-collapse. We found it hard to make an LLM judge correctly label "good" or "bad" Activation Oracle outputs, even with an explicit rubric. We also attempted GRPO-style RL with the following rubric:

- Passing the "swap test" (does the AO's answer change when given activations from a meaningfully different context?)
- Is the answer specific and falsifiable?
- Does the response add any meaningful insight?
- Is it not clearly, obviously wrong?
- Is the oracle following instructions?

This increased performance on some evals but caused regressions on others, and we are not confident the resulting checkpoint is more useful to researchers in practice. We are uncertain whether our implementation choices were optimal given time constraints. We nonetheless believe RL remains a promising direction for aligning AO outputs with desirable properties; the central bottleneck is getting an LLM judge to reliably discriminate good from bad AO responses.

A.5 Other differences compared to Karvonen et al. [2025]

- The number and identity of activation positions we feed is a random subset of the available positions (since our input is long). Adam's recipe uses "sometimes contiguous n, sometimes a single activation". We ablated this difference and did not find a significant change in eval performance.